



BIOLOGICAL DATA ANALYSIS WITH R

Exploratory Data Analysis & Visualization

Summarizing and Seeing the Shape of Biological Data

Hands-on session — summary stats · base plots · ggplot2 · heatmaps

Where We Left Off

Last session turned raw files into clean, analysis-ready biological data

1

Import

`read.csv()`, `read.table()` and `read_excel()` to bring data into R

2

Clean

`is.na()`, `duplicated()` and outlier checks before any analysis

3

Wrangle

`dplyr`'s `filter`, `mutate`, `group_by`, `summarise` and `left_join`

4

Reshape

`tidyr`'s `pivot_longer()` and `pivot_wider()` between wide and long formats

Today we explore that clean data — summarizing it with statistics and seeing its shape through base R plots, `ggplot2` and heatmaps

Learning Objectives

1

Summarize data numerically

Compute mean, median, mode, variance, SD and quartiles

2

Plot with base R

Build histograms, boxplots and scatterplots directly from a data frame

3

Apply the grammar of graphics

Build layered plots with ggplot2 — data, aesthetics, geoms

4

Visualize complex biological data

Facet multi-panel plots, and build heatmaps/correlation plots



```
> mean(df$od600); median(df$od600)
> var(df$od600); sd(df$od600)

> boxplot(od600 ~ treatment, data = df)

> library(ggplot2)
> ggplot(df, aes(treatment, od600)) +
+   geom_boxplot()

> heatmap(as.matrix(expr[, -1]))
```

Installing the Required Package

ggplot2 is not part of base R — install it once, then load it with `library()` every session. Everything else today uses functions already built into base R.

```
install.packages("ggplot2")
```

Adds the grammar-of-graphics plotting system used later in this session

```
hist() · boxplot() · plot() · heatmap() · cor()
```

Already built into base R — no installation needed for these



```
# Run once (skip if already installed)
install.packages("ggplot2")
```

```
# Run at the start of every session
library(ggplot2)
```

```
# Base R plotting/statistics need
# no library() call at all
hist(df$od600)
```

Summarizing Data Numerically

Before plotting anything, a handful of summary numbers describe the center, spread and shape of the data

`mean()` / `median()`

Central tendency — the average value and the middle value

`mode` (no built-in fn)

Most frequent value — use `table()` then `which.max()`

`var()` / `sd()`

Spread — variance and its square root, standard deviation

`quantile()` / `summary()`

Quartiles and a six-number summary (min, Q1–Q3, max, mean)

```
> df <- read.csv("culture_od600_clean.csv")
> mean(df$od600)
[1] 0.852
> median(df$od600)
[1] 0.58
> var(df$od600); sd(df$od600)
[1] 0.486
[1] 0.697
> summary(df$od600)
   Min.   1stQu.  Median   3rdQu.    Max.
0.214   0.393   0.581   0.967   2.224
```

Histograms, Boxplots & Scatterplots

Base R can produce a complete plot in a single function call — ideal for a quick first look at data

`hist(x)`

Shows the distribution/frequency of a single numeric variable

`boxplot(y ~ group)`

Compares the spread and median of y across groups

`plot(x, y)`

Draws a scatterplot showing the relationship between two variables

`par(mfrow = c(r,c))`

Arranges multiple plots together in one r-by-c grid

```
> hist(df$od600, main="OD600 Distribution",  
+       xlab = "OD600")  
  
> boxplot(od600 ~ treatment, data = df,  
+         col = c("lightblue","salmon"))  
  
> plot(df$day, df$od600,  
+       xlab = "Day", ylab = "OD600")
```

Introduction to ggplot2 & the Grammar of Graphics

Every ggplot2 plot is built the same way — data, then aesthetic mappings, then layers of geoms added with +

`ggplot(data, aes(...))`

Declares the dataset and maps variables to x, y, color, etc.

`geom_point()` / `geom_line()`

Adds a layer — points for a scatterplot, a line for a trend

`labs()` / `theme()`

Sets titles and axis labels; controls overall plot appearance

`+`

Chains layers together — each + adds one more element to the plot

```
> library(ggplot2)
> ggplot(df, aes(x = day, y = od600,
+   color = treatment)) +
+   geom_point(size = 2) +
+   geom_line(aes(group = sample_id)) +
+   labs(title = "Growth Over Time",
+     x = "Day", y = "OD600")
```

Visualizing Gene Expression & Growth-Curve Data

gene_expression.txt holds expression values for 8 genes across 4 samples — ideal for a heatmap and a correlation matrix

as.matrix()

Converts a numeric data frame into a matrix for heatmap()/cor()

heatmap()

Base R function — draws a color-coded matrix, clustering rows/columns

cor()

Computes pairwise correlation coefficients between numeric columns

ggplot2 + geom_tile()

Builds a custom heatmap layer-by-layer from long-format data

```
> expr <- read.table("gene_expression.txt",
+   header = TRUE, sep = "\t")
> mat <- as.matrix(expr[, -1])
> rownames(mat) <- expr$gene_id

> heatmap(mat, scale = "row")

> cor(mat)

               control_rep1 ...
control_rep1      1.00 ...
```

Key Takeaways

Descriptive Stats

`mean()` · `median()` · `var()` · `sd()` · `quantile()` — the numbers behind the plot

Base R Plots

`hist()` · `boxplot()` · `plot()` — fast, one-line exploratory plots

ggplot2

`ggplot()` + `aes()` + `geom_*()` + `facet_wrap()` — layered, publication-style plots

Heatmaps & Correlation

`heatmap()` · `cor()` — spotting patterns across genes and samples

Remember: always look at a summary number and a plot together — statistics can hide shape (like bimodal data), and plots alone can hide precise values.